

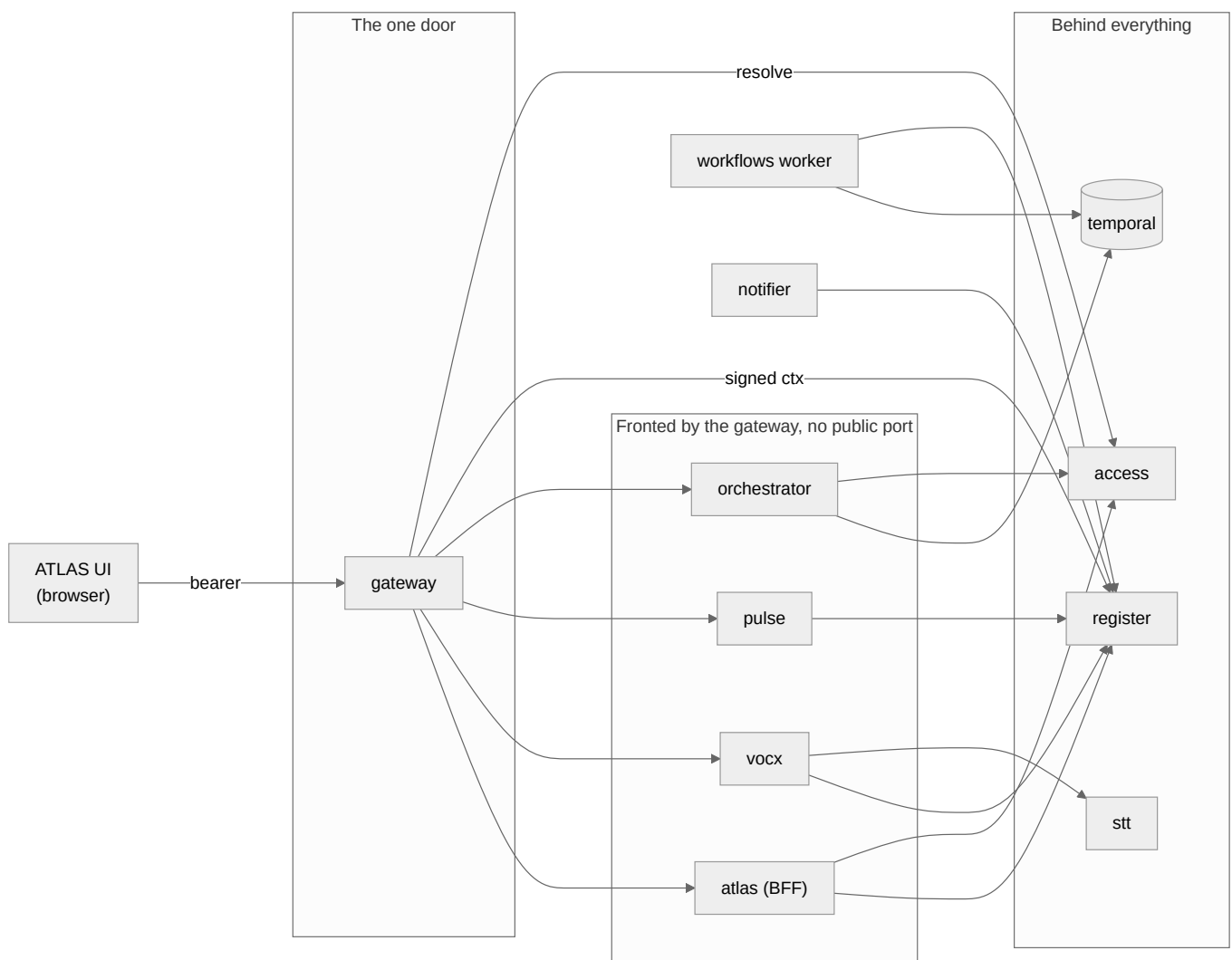
03 — Module Interaction

Audience: engineers changing anything that crosses a service boundary; security reviewers.

Companion docs: [01 Architecture](#) · [04 Running flows](#) · [07 RBAC](#)

This document answers three questions precisely: **who calls whom**, **what identity travels with the call**, and **what happens when the callee is down**.

1. The interaction map



Which credential each of those edges carries, and what happens when the callee is down, is the table in §2 — the labels are left off here so the shape stays readable.

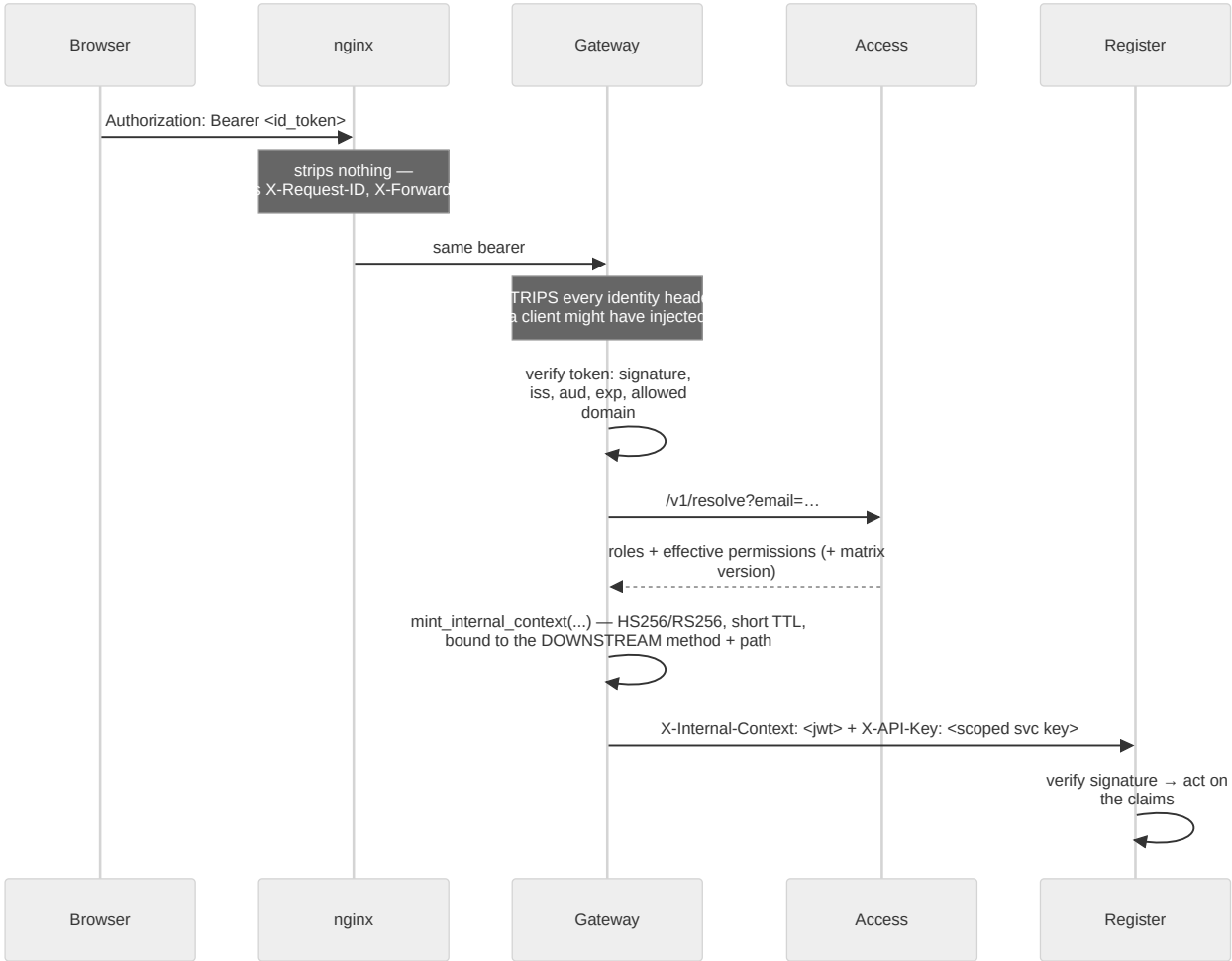
There are no other edges. In particular: the browser never talks to anything but the gateway (and the static UI bundle); no service calls another service's *database*; and nothing calls back into the gateway.

2. Call directory

Caller	Callee	Transport	Identity carried	If callee is down
Browser	gateway	HTTPS via nginx	OIDC bearer	UI shows an error; nothing is written
gateway	access	HTTP <code>/v1/resolve</code>	service key	Cached grant is reused (TTL); stale-cache fallback keeps the platform readable
gateway	register	HTTP	<code>X-Internal-Context</code> (signed) + scoped service key	502 with a problem envelope
gateway	atlas / vocx / pulse / orchestrator / access	HTTP, prefix stripped	signed context + that service's own injected key	502
atlas	register	HTTP (register-client SDK)	its own service key + <code>X-Tenant</code> , <code>X-User-Email</code>	Dashboard call fails; nothing else affected
atlas	access	HTTP <code>/v1/resolve</code>	service key	Cached permissions; stale-cache fallback
vocx	stt	HTTP multipart (OpenAI-compatible)	<code>VOCX_STT_API_KEY</code>	Retries within a budget, then a 504 the UI explains
vocx	register	HTTP	<code>svc_vox</code> key + <code>X-On-Behalf-Of</code>	Capture is archived; the write is reported as failed
pulse	register	HTTP with <code>Idempotency-Key</code>	<code>svc_pulse</code> key	Scan reports failure; a re-run is safe
orchestrator	temporal	gRPC	—	Workflow start fails loudly
orchestrator	access	HTTP	service key	Fails closed for authority rechecks
workflows worker	register	HTTP (activities)	<code>svc_workflows</code> key	Temporal retries the activity with backoff
notifier	register	HTTP	service key	Outbox rows stay unsent; retried next tick

3. How identity travels

This is the single most important thing to understand before changing anything at a boundary.



Headers the gateway strips from every inbound request

From `services/gateway/app/main.py`:

```
_SKIP_REQUEST_HEADERS = {
    "host", "content-length", "connection", "keep-alive", "transfer-encoding",
    "upgrade", "expect",
    "x-authz-decision", "x-gateway-auth", "x-user-email", "x-user-id",
    "x-user-roles", "x-user-report-ids", "x-user-reports", "x-internal-context",
    "x-api-key",          # the client never presents a backend data-plane key
    "x-admin-key",       # injected by the gateway for a verified Admin only
    "x-on-behalf-of",    # a claim only a NAMED SERVICE may make
}
```

Each entry is a specific attack closed:

Stripped header	What forwarding it would allow
<code>x-user-email</code> , <code>x-user-roles</code> , <code>x-user-id</code>	Impersonating anyone, including Admin
<code>x-authz-decision</code>	Declaring your own request FULL-authorized
<code>x-gateway-auth</code> , <code>x-internal-context</code>	Forging the trusted internal identity
<code>x-api-key</code>	Presenting a backend data-plane credential directly
<code>x-admin-key</code>	Reaching tenant administration without being Admin
<code>x-on-behalf-of</code>	Filing rows under a colleague's name from a browser

If you add an identity- or authorization-bearing header anywhere, add it to this set in the same commit. A header the gateway stamps but does not strip is a forgery channel.

The signed internal context

`packages/evam-backend-core/evam_backend_core/internal_token.py`. It replaced plaintext `X-User-*` headers plus a static shared secret, because:

- **Tamper-evident** — identity, roles *and* the effective grant are covered by the signature. A compromised downstream component cannot rewrite `X-User-Roles`.
- **Per-request and expiring** — a stolen token dies in `t1` seconds; a static secret never does.
- **Single source of truth** — the token carries the *live* effective matrix with its version, so the Register enforces exactly what Access resolved, not a stale local copy.
- **Replay-bound** — minted against the *downstream* (prefix-stripped) method and path, so it cannot be replayed on a different route.

Two signing modes: **HS256** (shared secret; fine inside one trust boundary) and **RS256** (gateway holds the private key, Register only the public one — use when the Register must be provably unable to mint gateway tokens).

Machine identity

Service-to-service calls that do *not* originate from a human use a **named** service API key. `packages/evam-backend-core/evam_backend_core/service_policy.py` maps each name to an explicit capability allowlist. `svc_pulse` gets exactly `{run_news_scan, edit_intel}`. This is code, not configuration, so widening it is a reviewed change.

4. Routing: how one door fronts everything

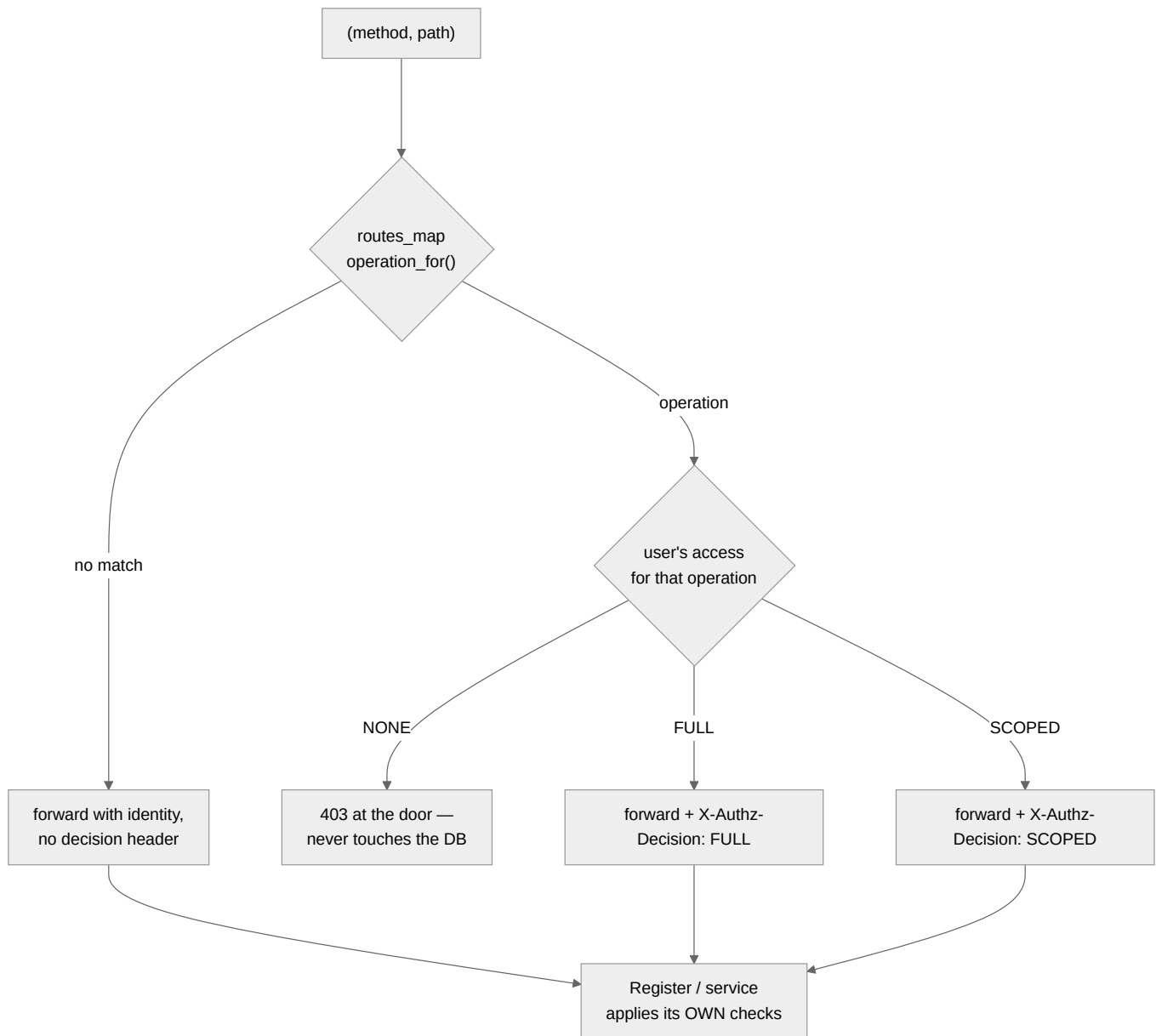
`services/gateway/app/main.py::_route()`:

```
prefixes = (
    ("/access",      settings.access_url,      settings.access_api_key),
    ("/atlas",       settings.atlas_url,       settings.atlas_api_key),
    ("/vocx",        settings.vocx_url,        settings.vocx_api_key),
    ("/pulse",       settings.pulse_url,       settings.pulse_api_key),
    ("/orchestrator", settings.orchestrator_url, settings.orchestrator_api_key),
)
# ...anything else → the Register
```

Two properties worth knowing:

1. **The prefix is stripped.** VocX sees `/v1/capture`, not `/vocx/v1/capture`. This is why `routes_map.py` entries for fronted services *include* the prefix — the gate runs before the strip.
2. **An unconfigured prefix falls through to the Register.** A partial deployment still works, and nothing becomes reachable around the gateway.

The edge gate is an accelerator, not the only gate



From the module docstring of `routes_map.py`: "A route NOT in this map forwards with identity headers but no decision; the Register then applies its own checks. Grow this map as routes are classified — unmapped is safe (enforced downstream), mapped is fast (rejected at the door)."

`SCOPED` is not an answer, it is a question passed along: the Register decides which rows this user may write, using the central scope evaluator in `evam_backend_core/policy.py` against line assignments.

5. Timeouts must agree across the chain

A hop with a shorter budget than the hop below it turns a working request into a 504. The three chains that matter:

Voice capture

Hop	Budget	Source
Browser (axios)	300 s	<code>CAPTURE_TIMEOUT_MS</code> in <code>services/atlas/ui/src/api/vocxClient.ts</code>
nginx <code>/vocx/v1/capture</code>	305 s	<code>deploy/nginx/nginx.conf</code>
gateway slow path	600 s	<code>_SLOW_PATHS</code> + <code>slow_upstream_timeout_s</code>
VocX → STT total budget	240 s	<code>stt.api.budget_s</code>

Shortest wins, so the browser decides. That ordering is intentional: the user's own client gives up first, with a message it can explain, rather than the edge returning an opaque 504.

CAM generation and the PULSE sweep

Hop	Budget
Browser	620 s
nginx <code>/orchestrator/v1/cam/</code> , <code>/pulse/v1/news/sweep</code>	625 s
gateway <code>_SLOW_PATHS</code>	600 s

Everything else

65 s at the edge, 60 s at the gateway. A slow Register call is a fault and should fail fast.

Rule: if you add a legitimately slow endpoint, you must widen **three** places — nginx (`deploy/nginx/nginx.conf`), the gateway (`_SLOW_PATHS`), and the browser client — plus `gateway.ingress.slowPaths` for Helm. Widening one does nothing.

6. Failure behaviour, by design

Dependency	Posture	Rationale
Access down, normal requests	Fail open on cache — the last resolved grant is reused	A brief identity outage must not stop the desk reading its own book
Access down, sensitive operations (delete/restore, assignments, governed imports, evidence break-glass) with <code>REGISTER_ONLINE_REVALIDATION=true</code>	Fail closed (503)	A revocation must take effect immediately for actions that cannot be undone
Register down	Hard failure everywhere	It is the book; there is nothing sensible to serve
STT down / slow	Retry within a budget, then 504 with a plain-language message; the audio is already archived	The user must never lose a recording they made
Temporal down	Workflow starts fail loudly; in-flight workflows resume when it returns	Durable by construction
PULSE sources down	Scan reports partial failure; nothing written	Intel is advisory
Google Workspace down	VocX capture still completes; the Drive/Docs write is skipped	The register write is the part that matters
MinIO down	Document upload fails; metadata is not written either	No dangling rows

7. Idempotency and duplicate suppression

Path	Mechanism
PULSE intel writes	<code>Idempotency-Key: pulse:{tenant}:{entity}:{hash}</code> — a re-run never duplicates an alert
Workflow activities that write	Explicit <code>idempotency_key</code> argument on the activity, backed by the <code>idempotency_keys</code> table
Lead conversion	Transactional endpoint plus an idempotency key; a retry cannot create a second deal
VocX capture	<code>capture_id</code> stamped on the extraction and carried through
Register generally	<code>idempotency_keys</code> table; <code>POST</code> routes that matter accept the header

Known gap: `POST /v1/{subject}/{id}/interactions` is not idempotency-keyed. A retried interaction write can produce a duplicate timeline row. This has been observed as a *display* duplicate caused by a UI bug (since fixed), not by duplicated writes — but the gap in the API is real and worth closing.

8. Changing a boundary — a checklist

When you add or move an endpoint that crosses services:

- ☐ Does the gateway route it? (`_route` prefixes, or it falls through to the Register.)
- ☐ Should it be gated at the door? (Add to `routes_map.py` — unmapped is safe, mapped is fast.)
- ☐ Is it slow? (nginx + `_SLOW_PATHS` + browser client + `gateway.ingress.slowPaths`.)
- ☐ Does it introduce a new header carrying identity or authority? (Add to `_SKIP_REQUEST_HEADERS`.)
- ☐ Does a *machine* call it? (Add the capability to that principal in `service_policy.py`.)
- ☐ Is it a write that a retry could duplicate? (Take an `Idempotency-Key`.)
- ☐ What is the correct behaviour when the callee is down? (Fail open or closed — §6.)
- ☐ Does the UI need a new service module? (`services/atlas/ui/src/services/` — one per domain.)